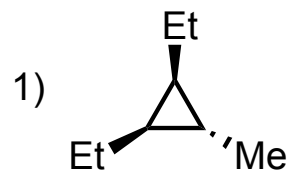
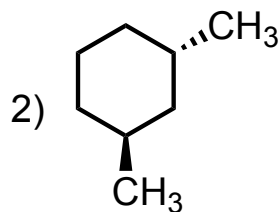


Identify each of the following molecule as chiral or achiral?

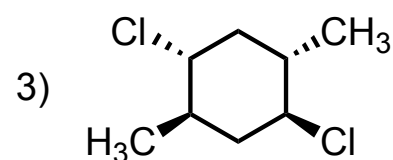


achiral

Plane of symmetry

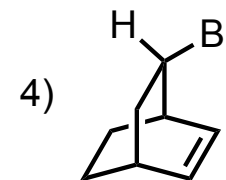


Chiral

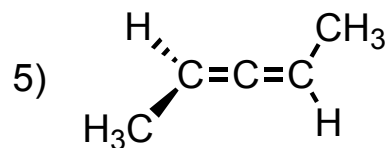


achiral

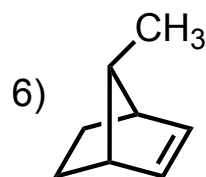
Centre of symmetry



Chiral

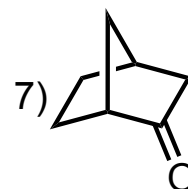


Chiral

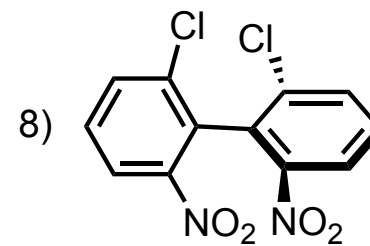


achiral

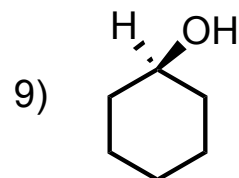
Plane of symmetry



Chiral

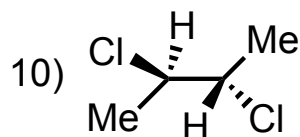


Chiral



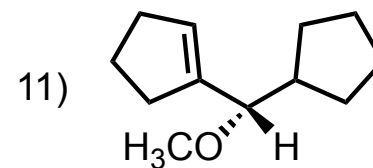
achiral

Plane of symmetry

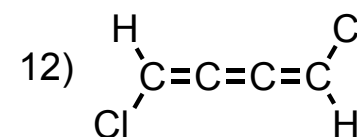


achiral

Centre of symmetry



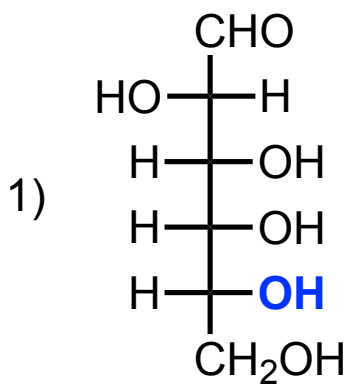
Chiral



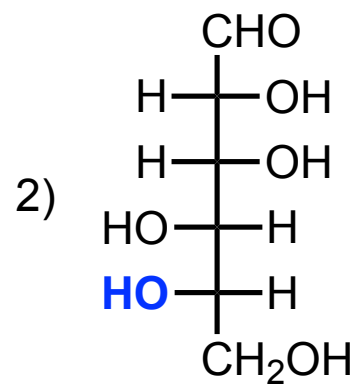
achiral

Centre of symmetry

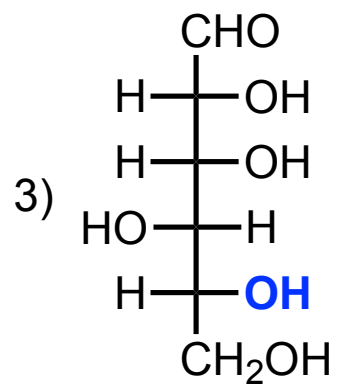
2) Assign D/L configuration for the following molecules.



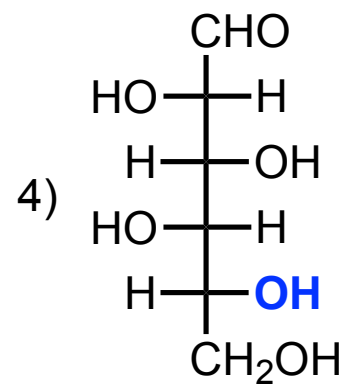
D - Altrose



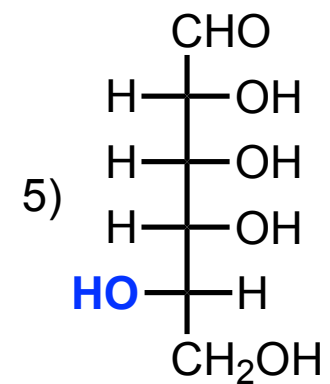
L - Mannose



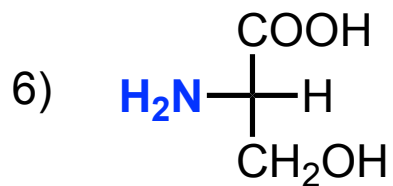
D - Glucose



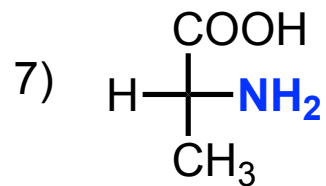
D - Idose



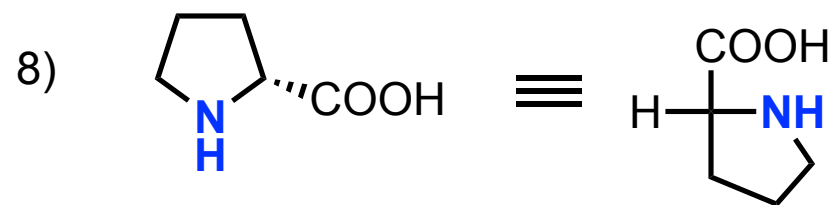
L - Talose



L - Serine

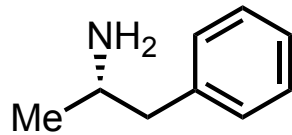


D - Alanine

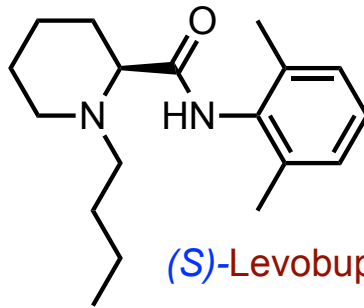


D - Proline

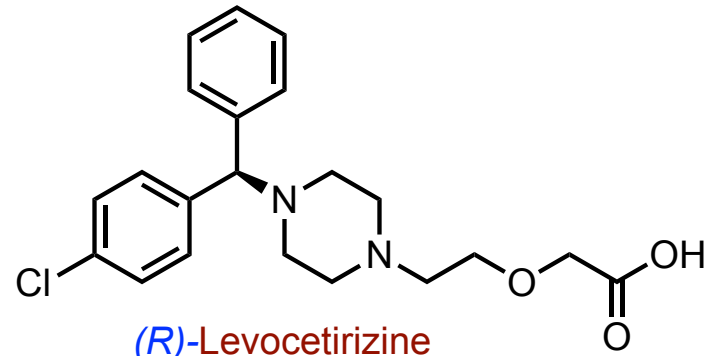
3) Assign R/S configuration for the following commercially selling medicinal compounds.



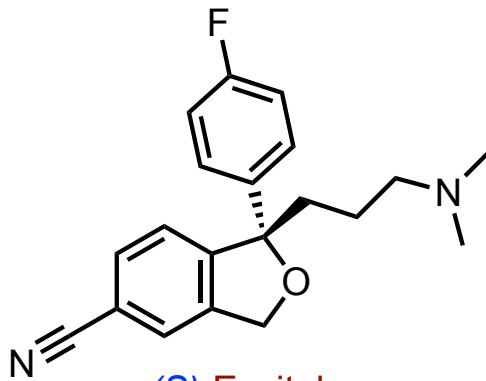
(S)-Dextroamphetamine



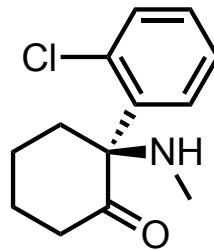
(S)-Levobupivacaine



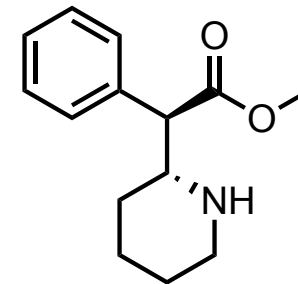
(R)-Levocetirizine



(S)-Escitalopram

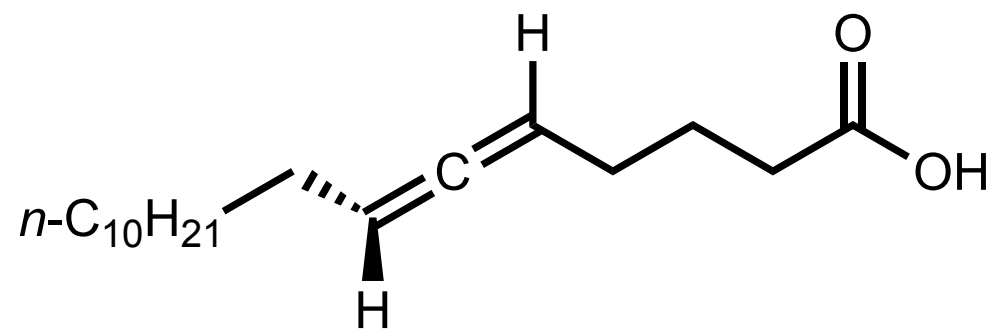


(S)-Esketamine

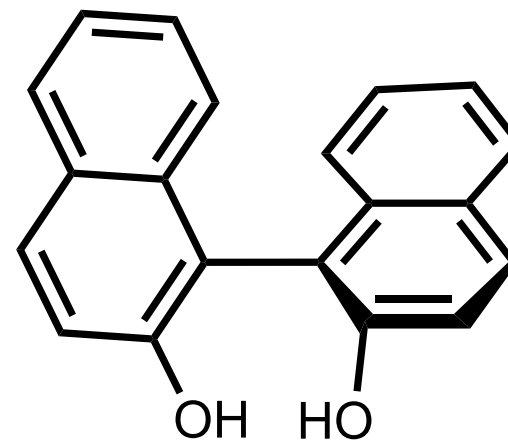


(R,R)-Dexmethylphenidate

4) Assign R/S configuration for the following compounds.

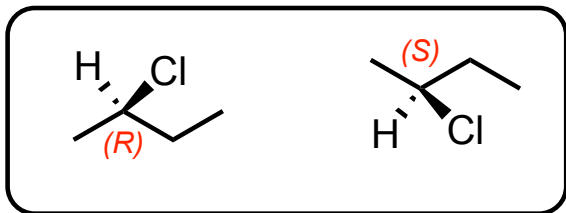


(R_a)-laballenic acid



(R_a)-BINOL

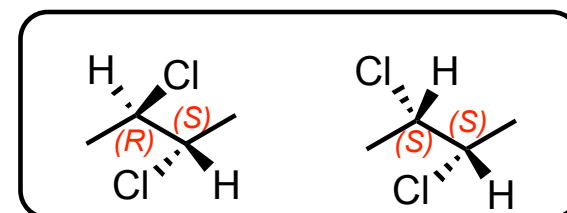
5) Identify the relationship between following stereoisomers (Enantiomers/Diastereomers/same molecules).



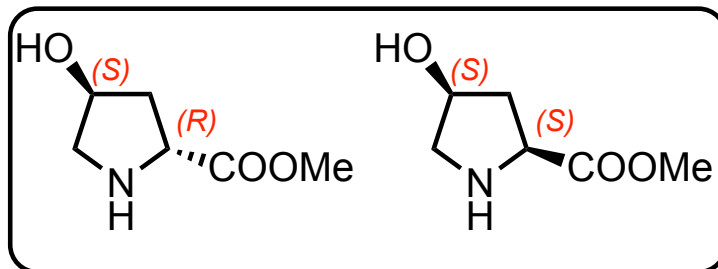
Enantiomers



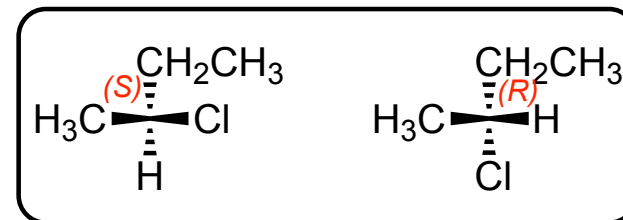
Same



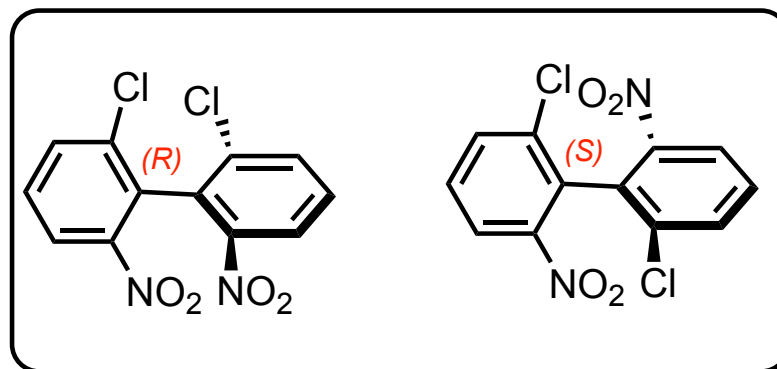
Diastereomers



Diastereomers



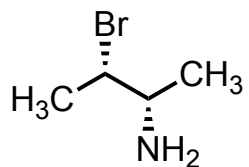
Enantiomers



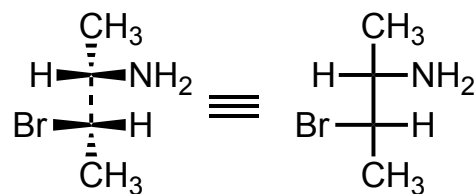
Enantiomers

6) Draw the Wedge, Fischer, Newman and Sawhorse projections for the following compounds

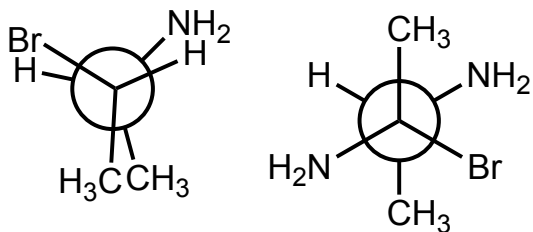
1) (2S,3S)-3-bromobutan-2-amine



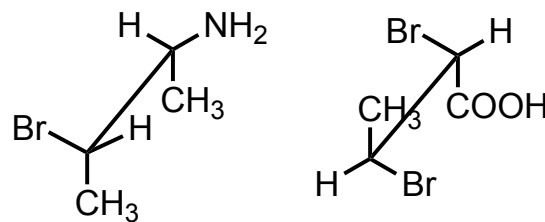
(2S,3S)-3-bromobutan-2-amine



(Fischer Projection)

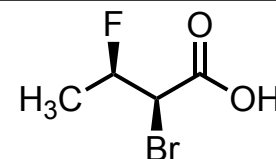


(Newman Projection)



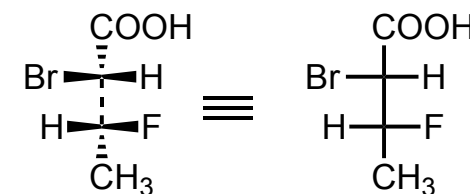
(Sawhorse Projection)

2) (2R,3R)-2-bromo-3-fluorobutanoic acid

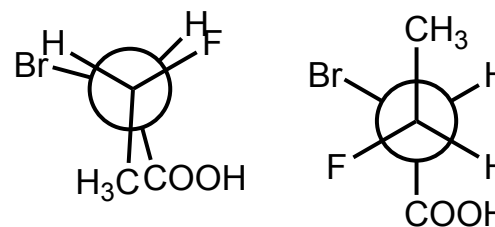


(2R,3R)-2-bromo-3-fluorobutanoic acid

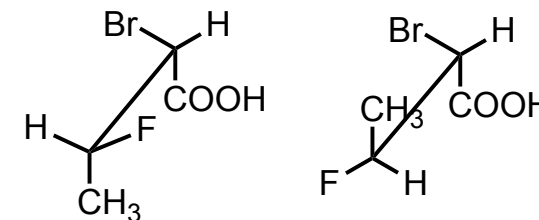
(Wedge Projection)



(Fischer Projection)

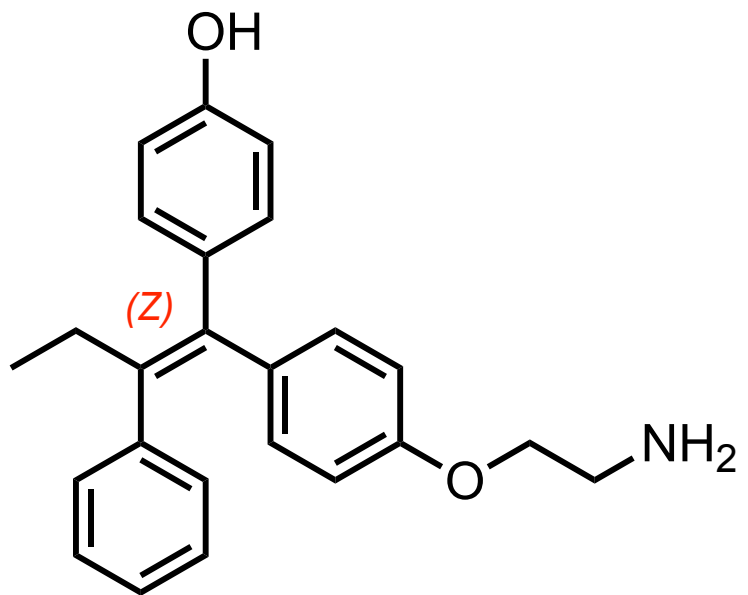


(Newman Projection)

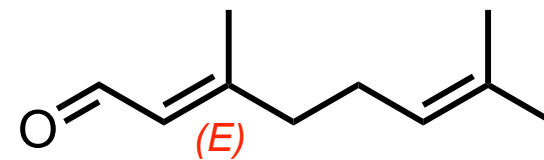


(Sawhorse Projection)

7) Assign E/Z configuration for the following molecules.



Norendoxifen



citral

8) A sample containing a single enantiomer of fluoxetine (Prozac) is placed in a sample tube (length = 12.5 cm) of polarimeter. The observed rotation is 9.06° clockwise at 25° C. The sample was made by dissolving 1.24 g of fluoxetine in a solution with a total volume of 2.62 mL. Calculate the specific rotation of the sample.

$$[\alpha] = \frac{\alpha}{c \times l}$$

$[\alpha]$ = Specific rotation

α = Observed rotation = +9.06°

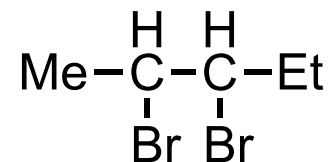
c = Concentration (g/ ml) = (1.24 g / 2.62 ml)

l = length (dm) = 12.5 cm = 1.25 dm

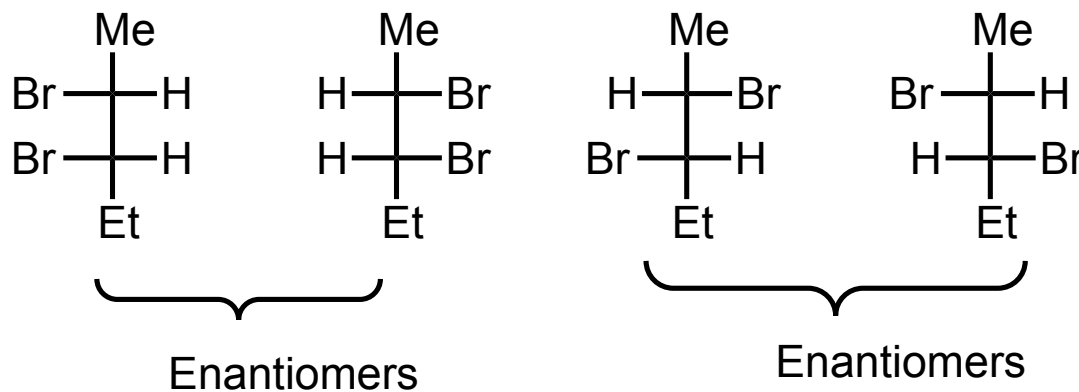
$$[\alpha] = [+ 9.06^\circ] / [(1.24 \text{ g} / 2.62 \text{ mL})(1.25 \text{ dm})]$$

$$[\alpha] = +15.3^\circ$$

9) Draw the possible number of stereoisomers for the following compound and also indicate the enantiomeric pairs .



Possible number of stereoisomers = 4



Possible number of Enantiomeric pairs = 2

10) How can you separate *rac*-2-pentanol using following chiral acid chloride?

